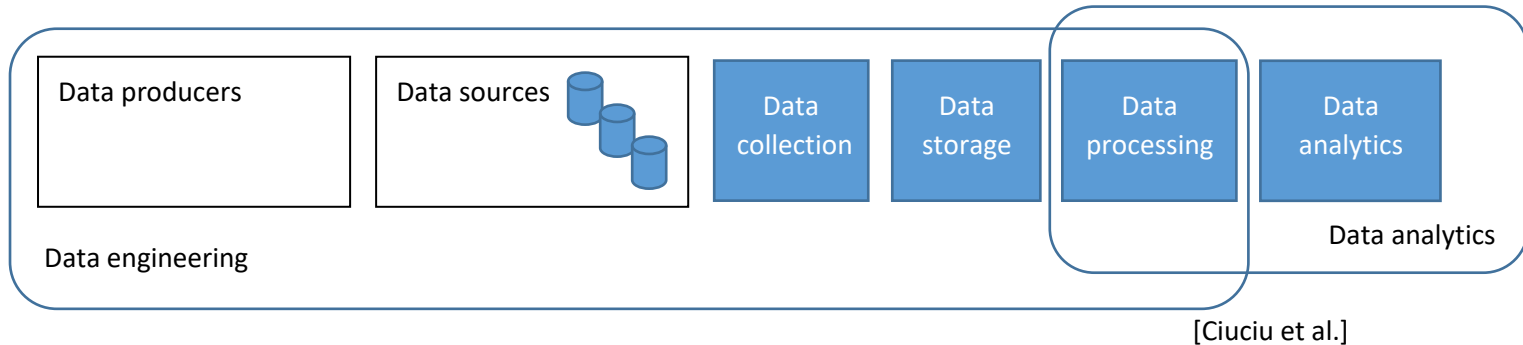


A standard architecture for big data/data science projects

Regardless the application, a data science project follows a standard pipeline:



Project description:

During the project the students are required to implement an end-to-end application with the following requirements:

1. Data Producers

- Any sensors or data producers (e.g., websites, mobile phones, cameras, etc.) that are necessary for the use-case

2. Data Sources

- In the case of a sensor, the data source are the sensor log files containing the sensor's parameters
- If external data is required for the use-case (e.g., public data) the relational database of the repositories where the data is stored could represent additional data sources

Remarks:

- At least 2 different data sources shall be used and integrated in the project

3. Data Storage

All data coming from the data sources should be stored in a scalable storage (e.g., NoSQL database) in json/avro/parquet format OR a similar solution (see cloud provider free solutions available). From here, the data will be prepared, aggregated and consumed by the algorithms/visualization dashboards in your application.

4. Data Processing

The project should consider and implement the batch (or, optionally, the real-time) processing layer(s) (or both). Simple aggregations on stored data are required before passing the data to the visualization dashboard.

5. The implementation of an algorithm

A ML (machine learning) model, statistical model which is proposed to solve the problem. The algorithms should be evaluated with proper statistical metrics in order to prove their validity.

Remark: recommender systems are also accepted.

6. Data visualization – requirement

- a. Visualization dashboard presenting the results of the algorithm in an easy to understand manner OR / AND
- b. Visualization dashboard for real time monitoring (e.g., the sensor parameters) – optional OR / AND
- c. Distributions of (sensor) parameters (e.g., histograms, bar plots) - batch and/or real-time (optional) visualizations.

Note: If the data are transmitted directly from the data source (sensor, data simulator) to the visualization tool, this is **not** considered data processing. You will need to make the data **go through a data processing pipeline** (batch or real-time or both) in your pipeline/solution.

Suggestion: follow the CRISP-DM process.

References:

1. I.Ciuciu, A.B. Ene, C. Lazar, An ICT Project Case Study from Education: A Technology Review for a Data Engineering Pipeline, Business Information Systems (BIS) 2019, LNBIP 353, ISBN: 978-3-030-20484-6, 2019